RESEARCH IDEA

Can one EEG channel classify sleep stages?

A reproducible deep-learning pipeline for automatic sleep staging

SINGLE-CHANNEL EEG

5 SLEEP STAGES

30-SECOND EPOCHS

A simpler input, a transparent evaluation, and a research-ready prototype.



Concept inspired by recent single-channel EEG sleep-staging research

The research idea is focused and testable

Reduce the input, preserve the signal information, and evaluate generalization.

CORE QUESTION

Can one channel support reliable sleep staging?

without relying on a
full multi-sensor PSG setup

PROJECT OBJECTIVES

01

Build

A single-channel EEG sleep-staging pipeline

02

Compare

Baseline CNN vs. the full representation + sequence model

03

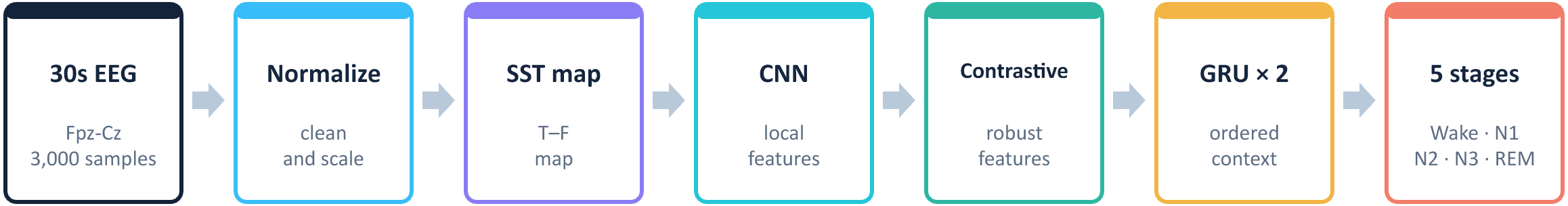
Evaluate

Class imbalance, unseen subjects, and per-stage performance

Research prototype — not a medical diagnostic device

The proposed workflow turns EEG into five labels

Each step has a specific role: normalize, represent, learn, and classify.



WHY THIS COMBINATION?

SST preserves time–frequency detail.
CNN learns local patterns. GRU preserves the order of feature windows.

PRIMARY METRIC

Macro-F1 + per-stage F1

because the classes are imbalanced

Model structure adapted from the selected recent research protocol

Data, evaluation, and expected contribution

The proposal ends with a reproducible experiment, not only a model score.

DATASET

Sleep-EDF Expanded

197 whole-night PSG recordings
EEG + expert-scored hypnograms

Input	Fpz-Cz EEG
Labels	Wake / N1 / N2 / N3 / REM
Split	subject-wise train / test

EVALUATION PLAN

01	**Baseline**	Majority class + CNN
02	**Ablation**	SST / contrastive / GRU / class weighting
03	**Metrics**	Accuracy, macro-F1, kappa, confusion matrix
04	**Generalization**	Unseen subjects, then external dataset

EXPECTED CONTRIBUTION: a transparent, reproducible single-channel EEG sleep-staging pipeline